

	1	Formatting and cleaning of longitudinal data (either in long or wide format and their interchangeability), Repeated measures and General linear model fitting, Model fitting for binary ordinal and count data (R, JAMOV, Mathematica, Stata, SAS)		
	Sections from References:			
Books and References:				
Diggle, P.J., Heagerty, P., Liang, K.Y and Zeger. S.L (2003). Analysis of Longitudinal Data- Second Edition. Oxford University Press, London.				
1. Fitzmaurice,M., Laird,M. and Ware, H. Applied Longitudinal Analysis- Second Edition. John Wiley & Sons, New Jersey.				
2. Crowder, M.J. and Hand, D.J. (1990). Analysis of Repeated Measures. Chapman and Hall/CRC Press, London.				
3. Hand, D and Crowder, M. (1996). Practical Longitudinal Data Analysis. Chapman and Hall/CRC Press, London.				
4. Lindsey, J.K. (1993) Models for Repeated Measurements. Oxford University Press, London.				
5. Little, R.J.A, and Rubin, O.B. (2019). Statistical Analysis with Missing Data- Third Edition. John Wiley & Sons, New York.				
6. McCullagh, P. and Nelder, J.A (1989). Generalized Linear Models- Second Edition. Chapman and Hall/CRC Press, London.				
7. Weiss, R.E. (2005). Modeling Longitudinal Data. Springer, New York				

SEMESTER VI

Programme	B. Sc. Statistics				
Course Code	STA6EJ301				
Course Title	Simulation Techniques				
Type of Course	Major Elective				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60

Pre-requisites	
Course Summary Objective	Statistical Methods to model and analyse a variety of Random Phenomena

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Generate pseudo-random numbers using different methods.	U	C	Instructor-created exams / Quiz
CO2	Use resampling methods on real datasets.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Apply Markov Chain Monte Carlo methods and density estimation	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Develop programs for simulation purposes.	U	C	Instructor-created exams / Home Assignments
CO5	Apply simulation skills in real-world scenarios	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I			10	15
	1	Introduction to random number generation.		
	2	Methods for generating random variables - Inverse transform		

		method		
	3	Composition method, Transformation method		
	4	Acceptance-Rejection method.		
	5	Generating from common statistical distributions Discrete and Continuous. (Rizzo (2019) and Rubinstein (2017		
	Sections from References:			
II			12	20
	6	Simulation for the multivariate normal distribution		
	7	Simple estimation based on simulated data		
	8	Monte Carlo integration and variance reduction techniques		
	9	Use of antithetic and control variables		
	10	Statistical validation of the simulated data by goodness of fit tests. (Rizzo (2019), Rubinstein (2017) and Ross (2022))		
	Sections from References:			
III			12	15
	11	Introduction to resampling,		
	12	Sampling distribution and other features of a statistic		
	13	Permutation and Randomization tests,		
	14	Theory for Jackknife, Variance estimation-consistency,		
	15	Jackknife in sample surveys,		
	16	Theory for the bootstrap and its consistency, Distribution and variance estimators (Shao & Tu (2012), Rizzo (2019))		
	Sections from References:			
IV			14	20
	17	Markov Chain Monte Carlo methods:		
	18	The Metropolis–Hasting’s algorithm		
	19	Gibbs sampling.		
	20	EM algorithm.		
	21	Smoothing with kernels		
	22	density estimation (McLachlan & Krishnan (1997), Rubinstein (2017),		